



AI Voice Receptionist for Hotels & Hospitality

Intelligent 24/7 multilingual hotel reception
powered by generative AI, live inventory, and
autonomous booking.

Group Name

Team Think

Team Members

Prabin Sharma
Nikesh Kumar Sah
Dhrub Narayan Yadav
Riyaj Khadka

Project

StayNep AI Voice Receptionist

Hotels get an always on multilingual front desk.

What the project is, who it serves, and why it matters.

One-sentence pitch

A voice-first AI receptionist that answers guest questions, checks availability, and completes bookings without front-desk handoff.

Problem

Hotels lose guest trust and bookings when staff are busy, offline, or unable to speak the guest's language.

Solution

StayNep combines speech, Gemini/OpenAI reasoning, hotel-specific knowledge, live inventory, and confirmations in one assistant.

Target users

Hotels, resorts, guesthouses, homestays, and travel operators that need affordable 24/7 guest service.

34

supported
languages

24/7

guest support

0

routine handoff
target

Routine hotel service still depends on staff availability.

The pain is most visible during off-hours, peak season, and multilingual guest interactions.

Off-hours

Guests still expect answers after reception desks slow down.

Language gaps

International and domestic travelers may not share one service language.

Missed bookings

Slow replies turn high-intent inquiries into lost revenue.

Staff load

Front desks repeat the same FAQ and booking tasks every day.

Real-life example: a guest asks about room availability late at night, wants a reply in their own language, and expects confirmation immediately. A traditional desk may need a callback; StayNep can continue the workflow in the same conversation.

The strongest signal is operational friction, not lack of websites.

Discovery focused on hotel owners, reception teams, guests, and tourism businesses.

Hotel owner

"Hiring and retaining reception staff is expensive and inconsistent."
Need: lower routine-service cost.

Reception team

"Peak-hour calls and repeated questions interrupt higher-value work."
Need: triage and automation.

Guest

"I want answers in my language without waiting for a person."
Need: instant, natural support.

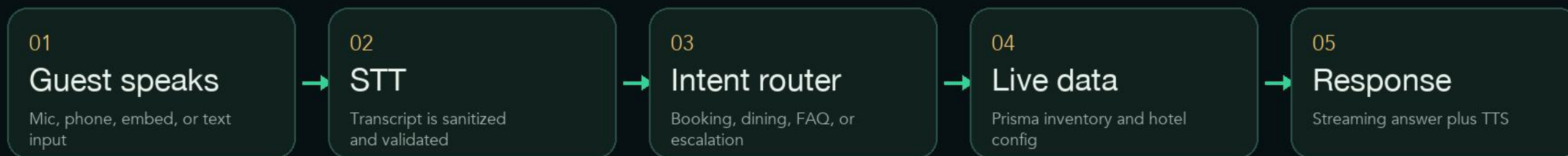
Tourism operator

"Slow inquiry follow-up can lose group bookings."
Need: faster conversion.

What we learned: guests do not want another static FAQ; hotels need an assistant that understands intent, checks real data, and knows when a human is truly required.

A complete guest service workflow runs inside the conversation.

Voice input becomes action: answer, availability check, booking, dining reservation, cancellation, or escalation.



Value

Guests get immediate service without waiting for reception.

Hotel benefit

Staff handle exceptions while AI handles routine volume.

Proof in repo

Booking, dining, RAG, guest auth, admin, and telephony routes are implemented.

StayNep moves from 'chatbot' to autonomous receptionist.

The Blue Ocean is voice + multilingual hospitality context + live operations.

Me Too: basic chatbot

- Text-only FAQ
- Scripted answers
- No inventory awareness
- No booking transaction
- Weak language switching

Traditional desk

- Human warmth
- Limited hours
- Single-location capacity
- High recurring cost
- Peak-hour overload

Only Me: StayNep

- Voice-to-voice service
- 34-language support
- Hotel-specific RAG
- Autonomous booking/dining
- FYI vs escalation routing

Competitive advantage: the assistant connects natural conversation to real hotel operations, so it can complete service rather than only describe service.

The product system covers guest, staff, and admin workflows.

Major capabilities visible in the application and codebase.

Voice-to-voice chat

STT, streaming
response, TTS

Autonomous booking

create, modify, cancel

Live inventory

availability and
alternatives

Dining reservations

venue, time, party
size

Guest accounts

My Stay panel and
history

Admin settings

hotel config, FAQ,
inventory

Staff notifications

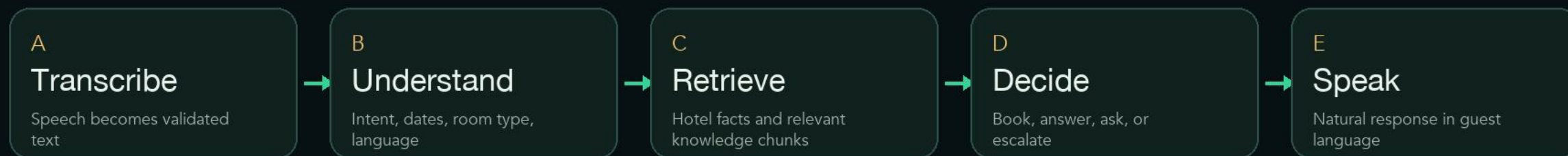
FYI emails and urgent
tickets

Embed/telephony

web widget, Telnyx,
WhatsApp

AI is the reasoning layer that turns speech into service.

Without AI, the product becomes a static form plus FAQ.



AI functionality

Gemini/OpenAI response engine, RAG augmentation, multilingual replies, booking-state reasoning, and escalation detection.

If AI is removed

The system loses natural language understanding, voice conversation, multilingual adaptability, and autonomous service decisions.

The architecture is modern, typed, and deployment ready.

Core technologies used by the current repository.

Frontend

Next.js 16
React 19
TypeScript
Tailwind CSS v4

Backend

App Router API routes
SSE streaming
Auth, rate limits
Validation

Database

Prisma ORM
PostgreSQL / Prisma
Postgres
Hotel, booking, guest
models

AI tools

Google Gemini
OpenAI fallback
RAG embeddings
Prompt engine

Voice/APIs

Nemotron/Whisper/Gemini STT
Edge/Nemotron/OpenAI TTS
Telnyx, WhatsApp, TingTing

Hosting

Vercel
Edge-ready routes
Service health checks

A subscription model converts hotel automation into recurring revenue.

Pricing can scale from small guesthouses to hotel groups.

Starter

\$49/mo

Includes

Small hotels
1 AI receptionist
Core FAQ + voice
Email support

Professional

\$149/mo

Includes

Mid-size hotels
Booking + dining flows
Analytics
Priority support

Enterprise

Custom

Includes

Hotel chains
Custom integrations
Telephony volume
Dedicated setup

Future sustainability: subscriptions, setup fees, PMS/CRM integration packages, and usage-based telephony or message volume.

Customers choose the option that combines availability, language, and action.

Simple comparison table across realistic alternatives.

Criteria	Human desk	FAQ bot	Generic AI tool	StayNep
24/7 service	Limited	Yes	Yes	Yes
Voice support	Yes	No	Partial	Yes
Hotel-specific data	Yes	Static	Manual setup	Built in
Live booking action	Manual	No	No	Yes
Multilingual scale	Limited	Low	Medium	34 languages
Staff workload	High	Medium	Medium	Lower

Market opportunity

Nepal's hotels and SMEs need affordable automation that feels local, multilingual, and operationally useful.

The project supports economic growth and digital innovation.

StayNep aligns most directly with SDG 8 and SDG 9.

SDG 8: Decent Work & Economic Growth

Reduces routine operational pressure.
Helps small hotels serve guests continuously.
Creates room for staff to focus on higher-value hospitality.
Supports tourism-driven business growth.

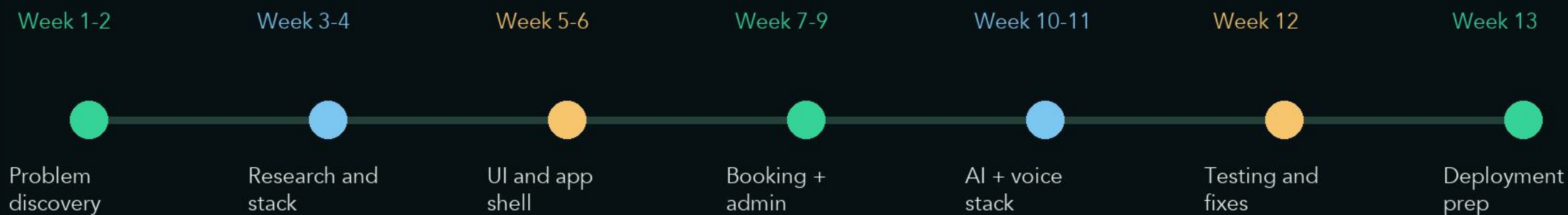
SDG 9: Industry, Innovation & Infrastructure

Applies generative AI to hospitality infrastructure.
Makes advanced automation accessible to SMEs.
Encourages local technology adoption.
Builds scalable digital service channels.

Impact: better guest service, stronger small-hotel competitiveness, and more efficient use of human staff.

The project moved from concept to working product through staged delivery.

Timeline summary for the Innovation Fest build.



Milestones achieved

Working voice assistant, autonomous booking flows, dining reservations, RAG knowledge, admin settings, analytics/support screens, guest auth, service health, and deployment configuration.

The hard parts were reliability, context, and production behavior.

Each challenge led to an implementation improvement.

Speech accuracy

Added multiple STT options, validation, and silence detection.

Hotel-specific answers

Built config-driven prompts and RAG knowledge chunks.

Booking conflicts

Used transactional booking service and alternative suggestions.

Escalation noise

Separated routine FYI notifications from urgent tickets.

Multi-tenant context

Introduced slug-based tenant routing and scoped config.

Deployment readiness

Added env examples, health route, tests, and provider fallbacks.

One more month would deepen integrations and reliability.

Roadmap from near-term demo polish to scalable product.



Scaling strategy

Package the assistant as an embeddable widget plus managed onboarding for hotels.

More AI features

Voice personalization, better multilingual evaluation, and smarter service recovery.

The demo should prove the core workflow in under five minutes.

Show login only if needed; move quickly to voice, booking, and output.



AI features to show

Speech-to-text, streamed response, TTS, multilingual reply, RAG hotel facts.

Operational output

Live availability, booking confirmation, staff FYI, guest My Stay panel.

Demo backup

Keep sample data ready, stable internet/hotspot, and a short recorded backup.

StayNep creates measurable value for hotels and guests.

The project solves a real service bottleneck with a working technical system.

24/7

continuous guest
service

34

language coverage

Live

inventory and
booking

FYI

not ticket spam

Problem solved

Guests get answers and
actions without
waiting for reception.

Value created

Hotels reduce routine
workload and recover
missed inquiries.

Innovation achieved

Voice AI is connected
to real hospitality
operations.

Every hotel deserves a receptionist that never sleeps.

Why users should adopt it, and why investors should believe in it.

Why users adopt

- Faster responses
- Service in the guest's language
- Booking without waiting
- Consistent answers from hotel data

Why hotels buy

- Lower routine workload
- More captured inquiries
- Scalable front desk capacity
- Clear escalation path

Why investors believe

- Subscription revenue
- Large hospitality SME market
- Expandable integrations
- Built product foundation

The future of hospitality is not only a person behind a desk. It is a service layer that is always available, multilingual, and connected to the hotel's real operations.

Team Think | StayNep AI Voice Receptionist